Math 420 - Exercise Set 1 (9.10.2026)

1. Let $X = \{a, b\}$ be a set. Write all possible topologies on X.

2. Let $X = \{a, b, c\}$ be a set. Write all possible topologies on X.

3. Consider all possible topologies on $X = \{a, b, c\}$. For each pair of topologies, determine if they are comparable.

4. Let X be a nonempty set and $\mathcal{T} = \{\emptyset\} \cup \{A \mid A \subseteq X, X \setminus A = A^c \text{ is countable}\}$ be a class of subsets of X. Show that $\mathcal{T}$ is a topology on X. ($\mathcal{T}$ is known as **cocountable topology or countable complement topology on X.**) Describe all $\mathcal{T}$-closed subsets of X.

5. Let $(X, \mathcal{D})$, $(X, \mathcal{T})$ and $(X, \mathcal{V})$ be a discrete space, cofinite topological space and cocountable topological space, respectively.

(a) If X is a finite set are $\mathcal{D}$, $\mathcal{T}$ and $\mathcal{V}$ coincident?

(b) If X is a countably infinite set do $\mathcal{D}$ and $\mathcal{V}$ coincide?

6. Let $A \subseteq X$ be any fixed subset of X. Show that $\mathcal{T} = \{\emptyset\} \cup \{U \mid U \subseteq X,\ A \subseteq U\}$ is a topology on X. Describe all $\mathcal{T}$-closed subsets of X. What topology results when $A = \emptyset$? When $A = X$?

7. Let $X = \mathbb{R}$ and $\mathcal{T} = \{\emptyset, X\} \cup \{(-\infty, a) \mid a \in \mathbb{R}\}$. Determine if $\mathcal{T}$ is a topology on X.

8. Let X be a set and x_0 be a fixed point in X. Show that $\mathcal{T} = \{\emptyset, X\} \cup \{U \mid U \subseteq X,\ x_0 \notin U\}$ is a topology on X. It is called the exluded point topology on X.

9. Give an example of topological spaces $(X, \mathcal{T})$, $(Y, \mathcal{V})$ and a $\mathcal{T}$-$\mathcal{V}$ continuous bijection $f : X \to Y$ such that f^{-1} is not continuous.

10. Give an example of topological spaces $(X, \mathcal{T})$, $(Y, \mathcal{V})$ and a $\mathcal{T}$-$\mathcal{V}$ continuous function $f : X \to Y$ such that f is not open.

11. Let $(X, \mathcal{T})$ and $(X, \mathcal{V})$ be two topological spaces, $\mathcal{B}$ and $\mathcal{N}$ be bases for $\mathcal{T}$ and $\mathcal{V}$, respectively. Show that $\mathcal{T} \subseteq \mathcal{V} \iff$ For every $B \in \mathcal{B}$ and every $x \in B$ there exists an $N \in \mathcal{N}$ s.t. $x \in N \subseteq B$.

12. Let $X = \mathbb{R}$ and $\mathcal{B} = \{[a, b) \mid a, b \in \mathbb{R},\ \ a < b\}$. Show that $\mathcal{B}$ is a base for some topology on $\mathbb{R}$. (This topology is called the lower-limit topology on $\mathbb{R}$.) Are the usual topology and the lower-limit topology on $\mathbb{R}$ comparable? If they are, is the usual topology on $\mathbb{R}$ coarser or finer than the lower-limit topology on $\mathbb{R}$?

13. Let $\mathcal{B}$ be a base for a topology $\mathcal{T}$ on a set X and let $\mathcal{B}'$ be a class of subsets of X with $\mathcal{B} \subseteq \mathcal{B}' \subseteq \mathcal{T}$. Show that $\mathcal{B}'$ is also a base for $\mathcal{T}$.

14. Let $(X, \mathcal{T})$, $(Y, \mathcal{V})$ be topological spaces and $f : X \to Y$ be a bijection . Show that if f is open and continuous then f is a homeomorphism.

15. Determine if each of the following classes of subsets of the plane $\mathbb{R}^2$ is a base for the usual topology on $\mathbb{R}^2$:

(i) The class of open rectangles whose sides are parallel to the coordinate axes,

(ii) The class of open equilateral triangles,

(iii) The class of open squares whose sides are parallel to the coordinate axes.